

SMART CITY COMPLAINT ANALYSIS

ABSTRACT:

The Smart City Complaint Analysis project focuses on analyzing citizen complaints related to roads, sanitation, water supply, traffic, electricity, and public health services. Using a dataset of 200,000 complaints, data cleaning, KPI creation, and trend analysis were performed to evaluate service efficiency and identify recurring urban issues. An interactive Power BI dashboard was developed to monitor complaint patterns, resolution performance, complaint hotspots, and citizen satisfaction. The insights generated help city authorities improve decision-making, optimize resource allocation, and enhance overall service delivery.

INTRODUCTION:

Modern cities receive thousands of citizen complaints every day, making it difficult for authorities to monitor and resolve issues efficiently. This project aims to analyze complaint data and identify patterns related to complaint categories, affected zones, department performance, and resolution delays. Using Python and Power BI, key performance indicators and interactive dashboards were developed to provide actionable insights. The analysis supports data-driven decision-making and helps city authorities improve urban services and citizen satisfaction.

DATASET DESCRIPTION:

The dataset contains 200,000 citizen complaints collected across multiple city zones and departments.

Key Features:

- Complaint ID
- Citizen ID
- Department
- Complaint Type
- Priority Level
- Status
- Complaint Date
- Resolution Days
- Area
- Zone
- Ward
- Severity Score
- Resolution Delay
- Satisfaction Score

DATA PREPARATION:

Handling Missing Values:

Missing values were found in four categorical columns:

- Department
- Priority
- Status
- Area

Missing entries in the Department, Priority, and Status columns were replaced using the most frequently occurring category in each column.

Missing values in the Area column were replaced with "Unknown" to retain all complaint records without losing information.

Handling Duplicate Records:

The dataset was checked for duplicate records to prevent inaccurate complaint counts and misleading insights. During the analysis, 10,000 duplicate records were identified.

These duplicate entries were removed to ensure that each complaint was represented only once in the dataset.

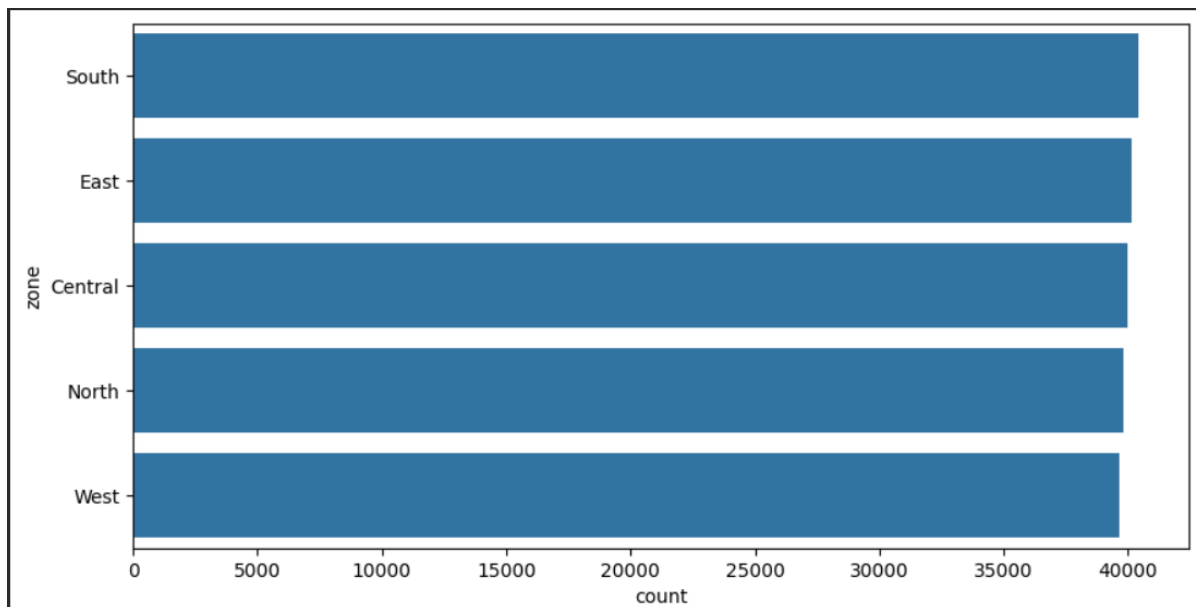
KPI SUMMARY:

KPI	Value
Total Complaints	2,00,000
Resolved Complaints	54,631
Resolution Rate	27.32%
Average Resolution Time	15.02 Days
High Priority Complaints	57,794
Average Citizen Satisfaction Score	2.90 / 5

ANALYTICS INVESTIGATION:

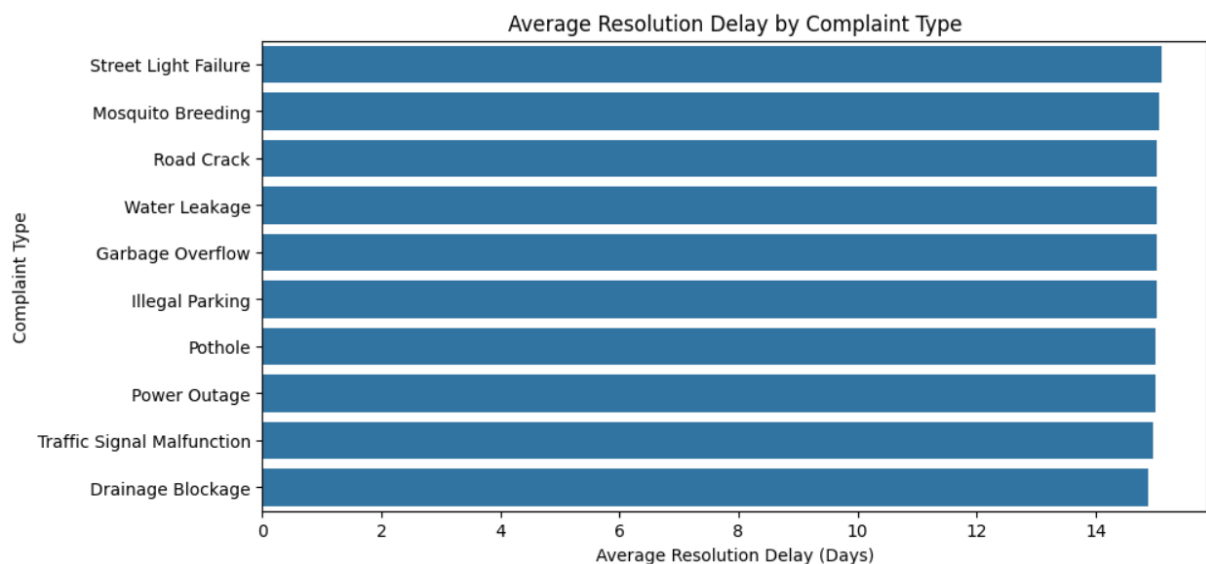
Zone vs Complaint Count:

Zone-wise analysis revealed that the South Zone recorded the highest number of complaints, followed by the East and Central Zones, indicating recurring civic issues in these areas.



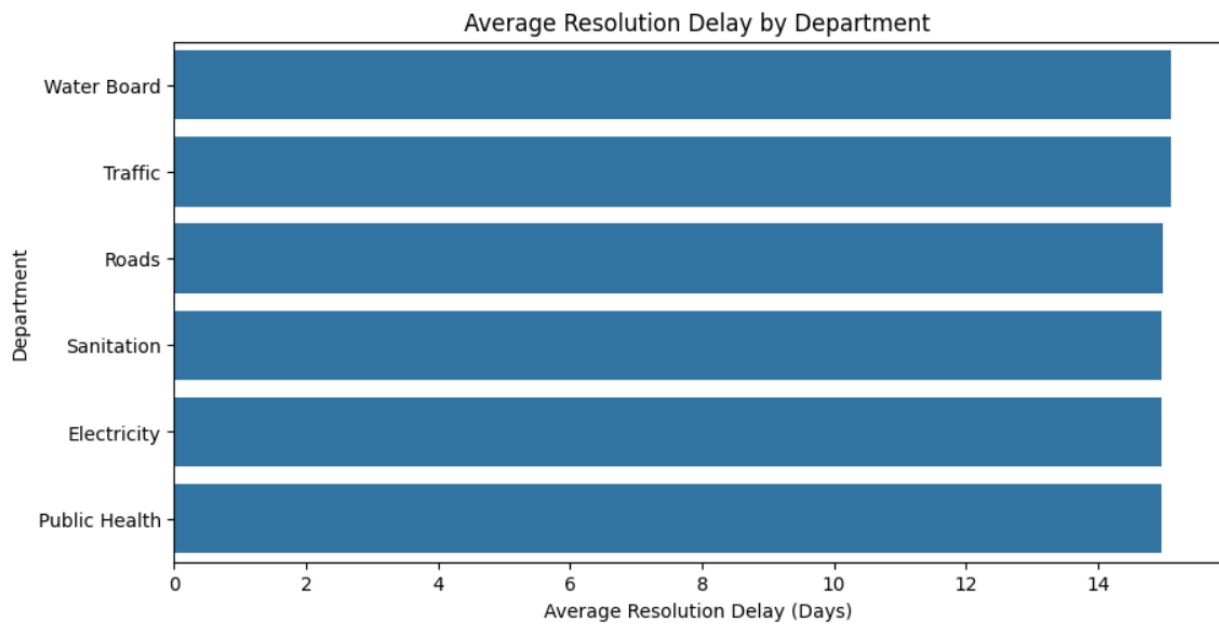
Category vs Resolution Time:

Traffic-related, drainage, and utility complaints required comparatively longer resolution times, highlighting areas that need operational improvements.



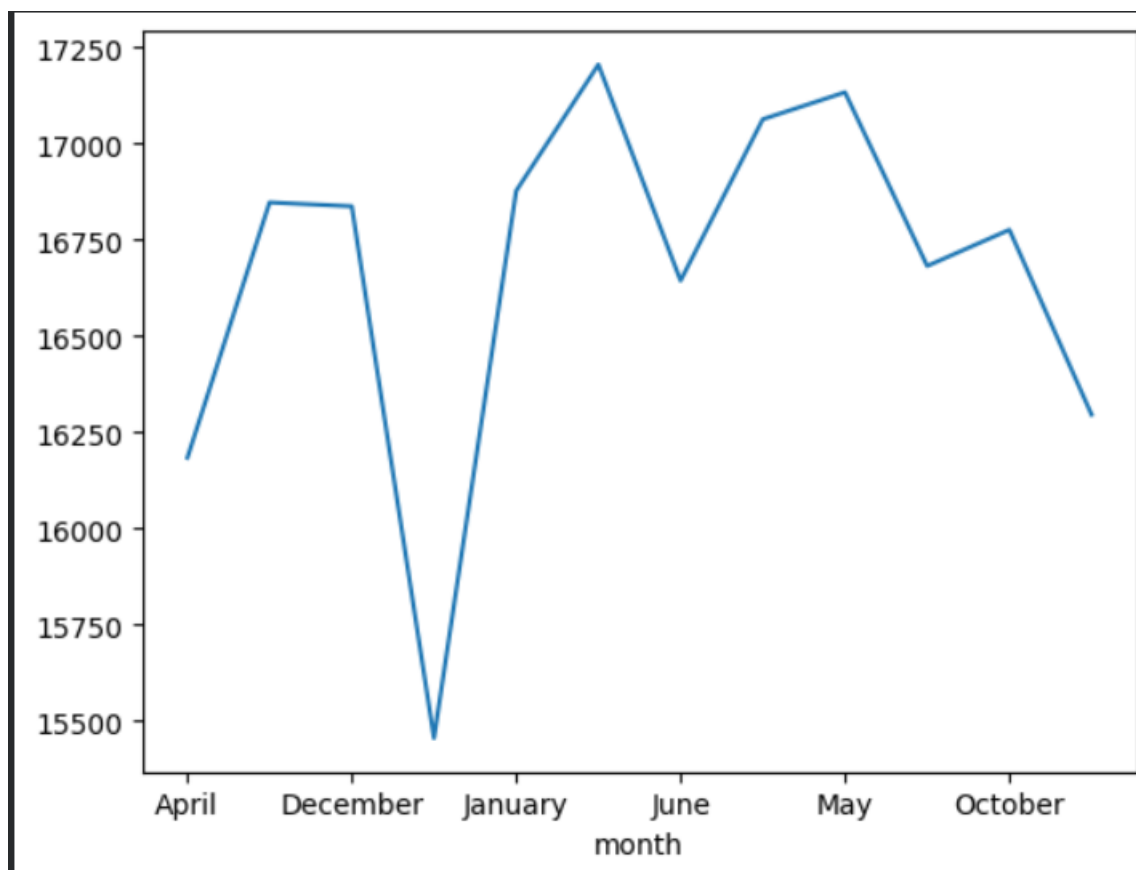
Department vs Performance:

The Water Board and Traffic departments showed the highest average resolution delays, suggesting heavier workloads and lower service efficiency.



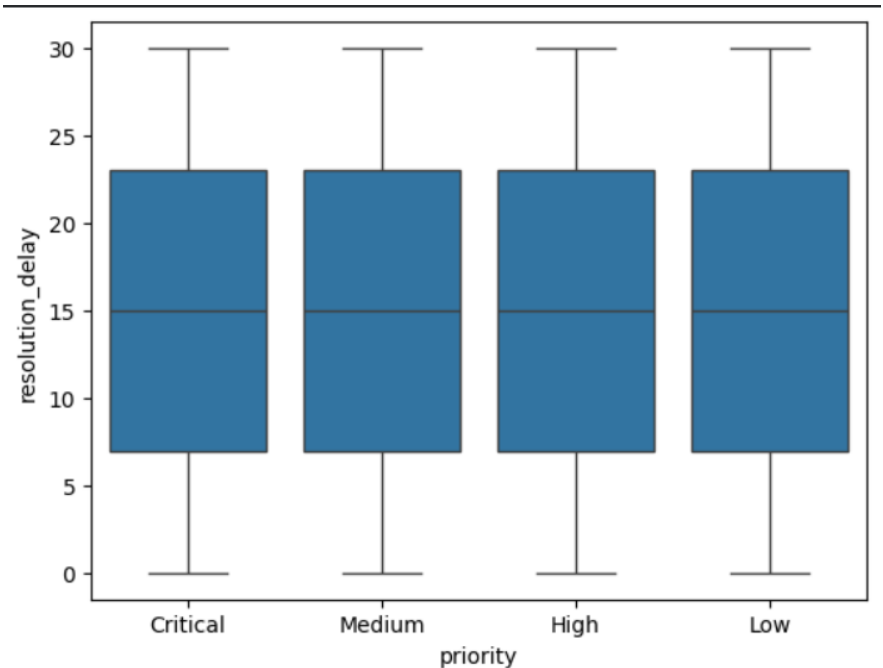
Complaint Trends Over Time:

Complaint trends showed periodic increases in complaint volume, indicating recurring infrastructure and service-related challenges.

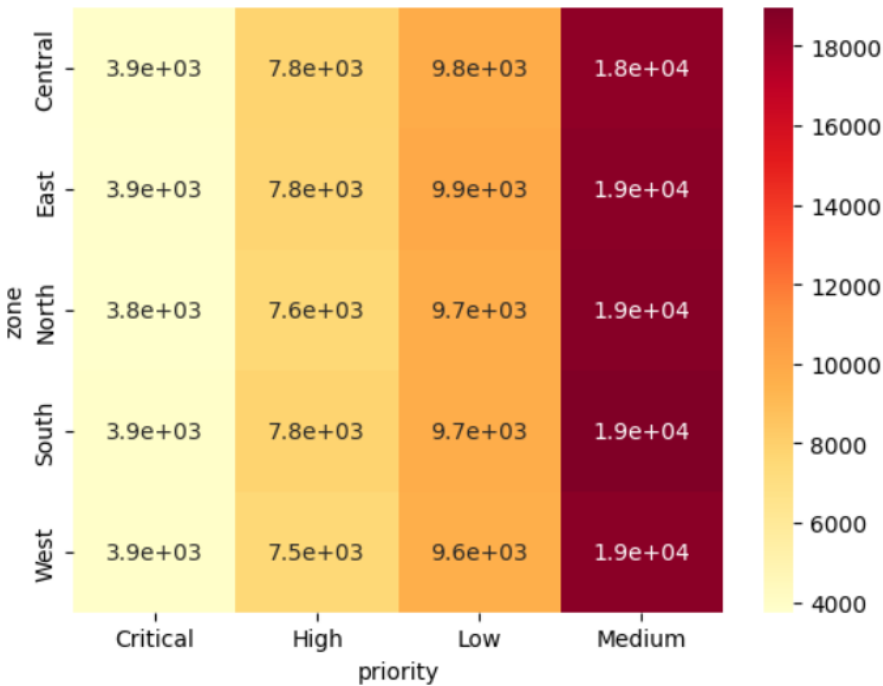


Priority vs Resolution Delay:

High-priority complaints experienced significant resolution delays, emphasizing the need for faster handling of urgent issues.



A heatmap was used to analyze the distribution of complaint categories across different zones. This visualization helped identify complaint hotspots, recurring civic issues, and high-impact areas requiring targeted intervention by city authorities.



Key Findings:

- 1. The South Zone is the primary complaint hotspot, followed by the East and Central Zones.
- 2. The Water Board and Traffic departments appear to be the most overloaded due to their higher resolution delays.
- 3. Traffic Signal Malfunction, Drainage Blockage, Power Outage, Street Light Failure, and Pothole complaints require immediate attention due to their high frequency and impact on citizens.

DASHBOARD DEVELOPMENT:

Executive Overview Dashboard:

The Executive Overview dashboard provides a high-level summary of complaint management performance across the city. It includes key KPIs such as Total Complaints, Resolution Rate, Citizen Satisfaction Score, and Complaint Trends. This dashboard helps decision-makers monitor overall service performance and identify major trends in citizen complaints.



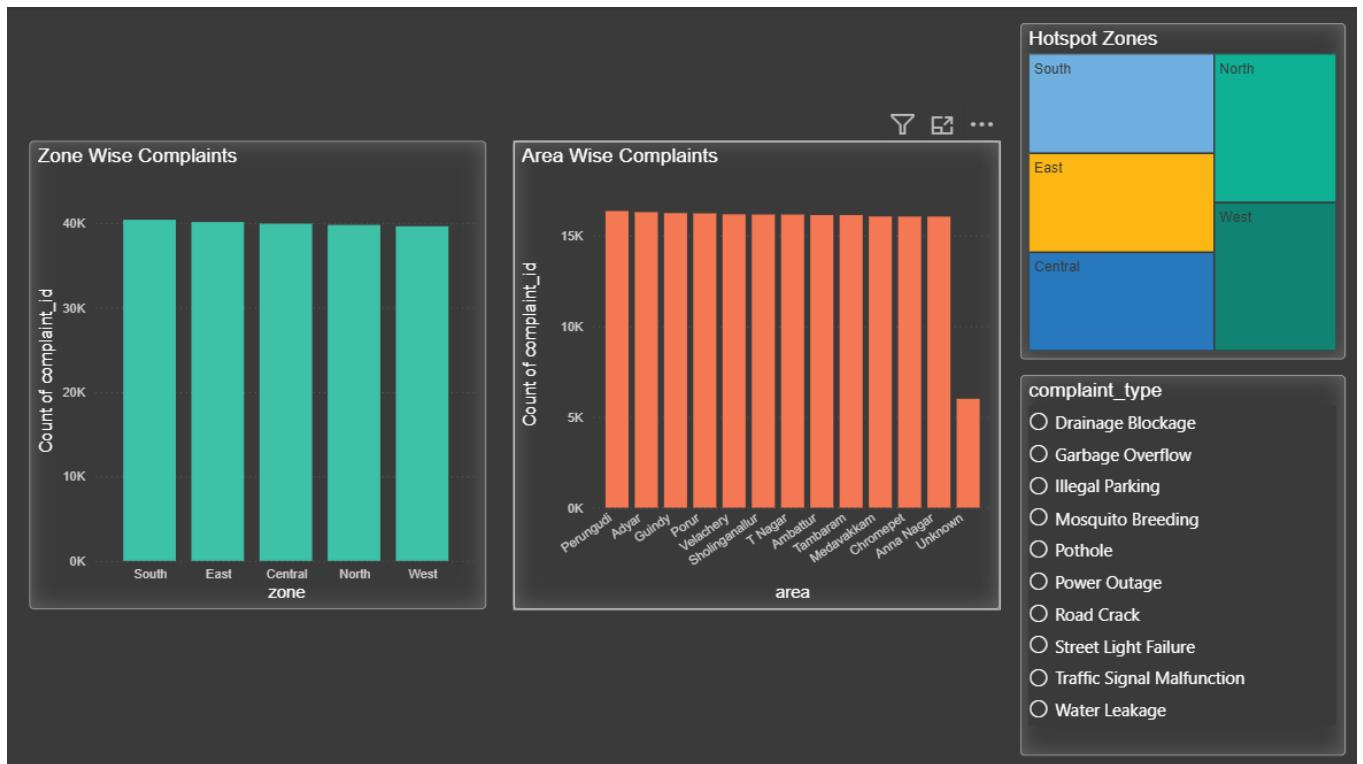
Complaint Dashboard:

The Complaint Dashboard focuses on complaint-related analysis by examining complaint categories, resolution patterns, and complaint growth trends. It helps identify the most common complaint types, understand how complaints are resolved, and monitor changes in complaint volume over time.



Geographic Dashboard:

The Geographic Dashboard provides a spatial view of complaints across different city zones. It analyzes zone-wise issues, identifies complaint hotspots, and visualizes the geographic distribution of complaints. This helps authorities locate recurring problem areas and prioritize regional interventions.



Service Performance Dashboard:

The Service Performance Dashboard evaluates the effectiveness of city departments in handling complaints. It analyzes department performance, resolution efficiency, and priority complaint handling. The dashboard helps identify overloaded departments, monitor service quality, and improve complaint resolution processes.



BUSINESS QUESTIONS:

1. Which complaint categories occur most frequently?

The most frequently reported complaint categories are:

1. Traffic Signal Malfunction – 20,138 complaints
2. Drainage Blockage – 20,083 complaints
3. Power Outage – 20,070 complaints
4. Street Light Failure – 20,037 complaints
5. Pothole – 20,030 complaints

Traffic infrastructure and drainage-related issues represent the largest share of complaints, indicating recurring urban maintenance challenges.

2. Which areas experience recurring civic problems?

Zone-wise complaint analysis shows:

Zone	Complaints
South	40,423
East	40,157
Central	39,957
North	39,808
West	39,655

South Zone records the highest complaint volume and can be considered the primary complaint hotspot. The distribution is relatively balanced across zones, suggesting city-wide infrastructure challenges rather than issues isolated to a single area.

3. Which departments take the longest to resolve complaints?

Average Resolution Delay by Department:

Department	Avg Delay (Days)
Water Board	15.12
Traffic	15.11
Roads	14.98
Sanitation	14.97
Electricity	14.97
Public Health	14.96

The Water Board and Traffic departments show the highest average resolution delays. These departments may require additional resources, automation, or process improvements to improve service efficiency.

4. What factors reduce citizen satisfaction?

Analysis of satisfaction scores and resolution delays reveals a strong relationship between service delays and citizen satisfaction.

Satisfaction Score	Avg Resolution Delay
5	1.50 Days
4	5.50 Days
3	11.51 Days
2	23.0 Days

Longer resolution delays significantly reduce satisfaction.

Complaints taking more than 20 days to resolve receive the lowest satisfaction ratings.

Unresolved and delayed complaints are the primary drivers of dissatisfaction.

High complaint volumes in hotspot zones further impact citizen perception of city services.

5. What actions should city authorities prioritize?

Immediate Actions

1. Reduce resolution delays in Water Board and Traffic departments.
2. Prioritize Traffic Signal and Drainage complaints.
3. Deploy rapid-response teams in South Zone.
4. Increase monitoring of high-priority complaints.
5. Implement complaint escalation for cases exceeding service targets.

Strategic Actions

1. Introduce predictive maintenance programs.
2. Develop real-time complaint monitoring dashboards.
3. Create departmental SLA targets.
4. Improve citizen feedback collection mechanisms.
5. Use AI-based forecasting to identify future complaint hotspots.

KEY FINDINGS:

- Traffic Signal Malfunction is the most common complaint category.
- South Zone is the largest complaint hotspot.
- Water Board has the highest average resolution delay.
- Resolution delay is the strongest factor affecting citizen satisfaction.
- Only 27.32% of complaints are marked as resolved, indicating substantial room for operational improvement.

RECOMMENDATIONS:

1. Increase resources for Water Board and Traffic departments.
2. Focus preventive maintenance efforts on traffic infrastructure and drainage systems.
3. Establish department-level performance monitoring.
4. Implement automated complaint prioritization.
5. Improve response times for high-severity complaints.
6. Continuously monitor citizen satisfaction metrics.

FUTURE SCOPE:

Future versions of this project can include:

- Real-time complaint monitoring
- GIS-based hotspot mapping
- Machine Learning complaint prediction
- Citizen sentiment analysis
- AI-powered complaint routing

- Automated service-level compliance monitoring

INNOVATION:

Citizen Satisfaction Segmentation can be introduced as an additional analytical layer to classify citizens based on satisfaction levels and evaluate the relationship between service performance and citizen experience. This segmentation helps transform complaint analytics into citizen-centric decision-making.

Segmentation Criteria

Satisfaction Score	Segment
1 – 2	Dissatisfied Citizens
3	Neutral Citizens
4 – 5	Satisfied Citizens

CONCLUSION:

The Smart City Complaint Analysis project analyzed 200,000 citizen complaints to identify complaint trends, service performance, and recurring civic issues. The analysis highlighted major complaint categories, complaint hotspot zones, and departments with higher resolution delays. Interactive dashboards and KPIs provided valuable insights into complaint management and citizen satisfaction. Overall, the project demonstrates how data analytics can help city authorities improve service efficiency, prioritize critical issues, and make data-driven decisions for better urban governance.